

Math 118 – Lecture 05

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1 Main idea

Just recall an adjoint $A : X \rightarrow Y$ which is a bounded linear operator and an adjoint is $A^* : Y \rightarrow X$ defined by requiring

$$\forall x, y, \langle Ax, y \rangle_Y = \langle x, A^*y \rangle_X$$

So basically we are learning about the existence of adjoint etc.

Show $A : X \rightarrow Y$ where X, Y are complete inner product spaces (Hilbert Spaces). has an *adjoint* A^* .

Proof. Given $y \in Y$, $l(x) = \langle Ax, y \rangle$ belongs to X' .

The Riesz Representation Theorem $\exists! v \in X$ such that $\forall l(x) = \langle x, v \rangle$ So

$$\forall x, \langle Ax, y \rangle = \langle x, A^*y \rangle \iff A^*y = v.$$

□

Claim. A^* is a linear operator.

Proof. Given $y, z \in Y$ and $\alpha, \beta \in \mathbb{C}$,

$$v = A^*(\alpha y + \beta z)$$

is the unique vector for which

$$\langle Ax, \alpha y + \beta z \rangle = \langle x, v \rangle \quad \forall x \in X$$

Use properties of the inner product on the LHS:

$$\overline{\alpha} \langle Ax, y \rangle + \overline{\beta} \langle Ax, z \rangle$$

Then we have

$$\overline{\alpha} \langle x, A^*y \rangle + \overline{\beta} \langle x, A^*z \rangle$$

which is

$$\langle x, \alpha A^*y + \beta A^*z \rangle$$

where v is unique by riesz. □

Properties of the Adjoint:

- $(A^*)^* = A$
- $(AB)^* = B^*A^*$
- $(A^{-1})^* = (A^*)^{-1}$
- $\text{range}(A)^\perp = \ker(A^*)$

Proof of $(A^*)^* = A$ We have $\forall x \in X, y \in Y, \langle A^*, x \rangle_X = \langle y, Ax \rangle_Y$

(Rewrite of our formula) It is the condition for either A or A^* to be the adjoint of the other.

$$\langle A^*y, x \rangle = \langle y, Ax \rangle$$

Conjugate both sides You get

$$\overline{\langle x, A^*y \rangle} = \overline{\langle Ax, y \rangle}$$

Claim. $\|A^*\| = M$ ($M = \|A\|$) First show $\|A^*\| \leq M$, for any $y \in Y$

$$\|A^*y\|^2 = \langle A^*y, A^*y \rangle = \langle y, AA^*y \rangle$$

Use Cauchy: $\langle y, AA^*y \rangle \leq \|y\| \cdot \|AA^*y\| \cdot \|A^*y\|$

If $A^*y \neq 0$, divide

$$\|A^*y\| \leq M\|y\|$$

which is true for every y , and the zero case is also true.

Use $(A^*)^* = A$ to conclude and basically want to show $\|A\| \leq \|A^*\|$ You can write it as

$$\|Ax\|^2 = \langle Ax, Ax \rangle = \langle x, A^*Ax \rangle \leq \|x\| \cdot \|A^*Ax\| \leq \|x\| \cdot \|A^*\| \cdot \|Ax\|$$

To proof that $\text{range}(A)^\perp = \ker(A^*)$

$$y \in \text{range}(A)^\perp \iff \forall x \in X, \langle Ax, y \rangle = 0$$

Which is analogous to

$$\forall x \in X, \langle x, A^*y \rangle = 0 \iff A^*y = 0$$

Write the one-sentence point of today's lecture here.

2 Projections Pt 2.

We define $P : V \rightarrow V$ to be a projection if P is bounded and $P^2 = P$ (as an operator), where P^2 means composition: $P^2(x) = P(P(x))$. So $P^2 = P$ says applying P a second time does nothing new

— think of projecting a point onto the floor: once you are on the floor, projecting again does not move you. (For example, $P(x, y) = (x, 0)$ satisfies $P(P(x, y)) = (x, 0) = P(x, y)$.)

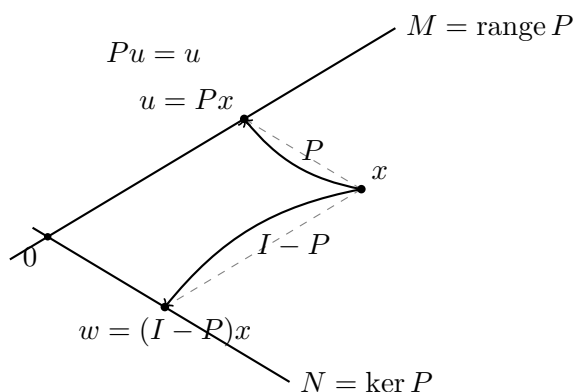
$I - P$ keeps whatever P threw away, so it should also be a projection. *Claim.* If P is a projection, so is $I - P$

Proof.

$$(I - P)^2 = (I - P - P + P^2) = (I - P) + (P^2 - P) = (I - P)$$

□

This is just the binomial expansion (legal since I commutes with everything), and the last step uses exactly the hypothesis: $P^2 - P = 0$.



So P keeps $\text{range } P = M$ and kills $\text{ker } P = N$, while $I - P$ does the reverse, splitting $V = \text{range } P \oplus \text{ker } P$ into the two complementary pieces.

Claim. For an orthogonal projection ($P^2 = P$ and $P^* = P$), we have $(I - P)u \perp Pv$ for all $u, v \in V$.

Proof. Using $\langle Ax, y \rangle = \langle x, A^*y \rangle$ with $P^* = P$,

$$\langle (I - P)u, Pv \rangle = \langle P(I - P)u, v \rangle = \langle (P - P^2)u, v \rangle = 0,$$

since $P(I - P) = P - P^2 = 0$.

□

This is exactly what “orthogonal” adds: a general projection only makes $\text{range } P$ and $\text{ker } P$ complementary (the two lines in the diagram, possibly skew); $P^* = P$ is what makes them perpendicular. Conversely, if $(I - P)u \perp Pv$ for all u, v , then taking $u \in \text{ker } P$ (so $(I - P)u = u$) gives $\text{ker } P \perp \text{range } P$, so P is the orthogonal projection onto its range.

Introduction: projecting without an orthonormal basis. We started from the orthonormal case: if $\{e_1, \dots, e_N\}$ is an orthonormal basis for V_0 , then

$$Pu = \sum_{n=1}^N \langle u, e_n \rangle e_n$$

is the orthogonal projection onto V_0 . Now let $\varphi_1, \dots, \varphi_N$ be a basis for V_0 that is *not* orthonormal. The coefficients are no longer just the inner products $\langle u, \varphi_n \rangle$ — so how do we project without running Gram–Schmidt?

The idea: since Pu lands in V_0 , it must be some combination

$$Pu = c_1\varphi_1 + c_2\varphi_2 + \dots + c_N\varphi_N$$

with unknown numbers $c_1, \dots, c_N$ — that is N unknowns, so we need N equations. The projection condition is that $u - Pu$ is perpendicular to every vector in V_0 , and it is enough to check it against the basis vectors:

$$\langle u - Pu, \varphi_n \rangle = 0 \quad \text{for } n = 1, \dots, N.$$

Expanding Pu and rearranging,

$$c_1 \langle \varphi_1, \varphi_n \rangle + c_2 \langle \varphi_2, \varphi_n \rangle + \dots + c_N \langle \varphi_N, \varphi_n \rangle = \langle u, \varphi_n \rangle, \quad n = 1, \dots, N.$$

That is an ordinary $N \times N$ linear system — solve it with Gaussian elimination, like any system from linear algebra. The matrix is all the pairwise inner products of the φ 's (it is called the *Gram matrix*); the right side is the inner products of u with each φ_n .

The Gram matrix, from scratch. Given vectors $\varphi_1, \dots, \varphi_N$, their *Gram matrix* G is the $N \times N$ table of all their pairwise inner products:

$$G_{mn} = \langle \varphi_m, \varphi_n \rangle, \quad m, n = 1, \dots, N.$$

That is the whole definition — row m , column n holds the inner product of φ_m with φ_n . What the table contains:

- **Diagonal entries:** $G_{nn} = \langle \varphi_n, \varphi_n \rangle = \|\varphi_n\|^2$ — the squared length of each basis vector.
- **Off-diagonal entries:** for real vectors, $\langle \varphi_m, \varphi_n \rangle = \|\varphi_m\| \|\varphi_n\| \cos \theta$ — they record the angle between φ_m and φ_n , and they are 0 exactly when the two vectors are perpendicular.
- $G = I$ exactly when the basis is orthonormal (all lengths 1, all angles 90°).
- G is always Hermitian: $G_{nm} = \overline{G_{mn}}$; for real vectors it is simply symmetric.

Two small examples. For $\varphi_1 = (1, 0)$, $\varphi_2 = (1, 1)$ in $\mathbb{R}^2$:

$$G = \begin{pmatrix} \langle \varphi_1, \varphi_1 \rangle & \langle \varphi_1, \varphi_2 \rangle \\ \langle \varphi_2, \varphi_1 \rangle & \langle \varphi_2, \varphi_2 \rangle \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

For the functions $1, t, t^2$ on $[0, 1]$, the entries are integrals — for example $\langle t, t^2 \rangle = \int_0^1 t \cdot t^2 dt = \frac{1}{4}$ and $\langle t^2, t^2 \rangle = \int_0^1 t^4 dt = \frac{1}{5}$ — which gives

$$G = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix}.$$

One identity worth remembering: expanding a squared norm with linearity,

$$\|c_1\varphi_1 + \cdots + c_N\varphi_N\|^2 = \sum_{m,n=1}^N c_m G_{mn} \overline{c_n}.$$

The Gram matrix is exactly the thing that turns a coefficient combination into a squared norm — which is why it is the coefficient matrix of the projection system, and why that system always has a unique solution when the φ 's form a basis: the right side above is a norm, so it is 0 only when every $c_n = 0$.

Sanity check: if the φ_n were orthonormal, the matrix would be the identity and we would get $c_n = \langle u, \varphi_n \rangle$ — exactly the formula we started from.

A tiny example. In $\mathbb{R}^3$, take $V_0 = \text{span}\{\varphi_1, \varphi_2\}$ with $\varphi_1 = (1, 0, 0)$, $\varphi_2 = (1, 1, 0)$ (this is the xy -plane, but the basis is not orthonormal), and $u = (1, 2, 3)$. The pairwise inner products are

$$\langle \varphi_1, \varphi_1 \rangle = 1, \quad \langle \varphi_1, \varphi_2 \rangle = 1, \quad \langle \varphi_2, \varphi_2 \rangle = 2,$$

and $\langle u, \varphi_1 \rangle = 1$, $\langle u, \varphi_2 \rangle = 3$. So the system reads

$$\begin{cases} c_1 + c_2 = 1 \\ c_1 + 2c_2 = 3 \end{cases} \implies c_2 = 2, c_1 = -1,$$

and $Pu = -\varphi_1 + 2\varphi_2 = (1, 2, 0)$ — the usual projection of $(1, 2, 3)$ onto the xy -plane, as expected. But notice $c_2 = 2$ while $\langle u, \varphi_2 \rangle = 3$: with a non-orthonormal basis the coefficients are *not* the inner products.

Functions work the same way. If V_0 is spanned by functions, say $\varphi_1 = 1$, $\varphi_2 = t$, $\varphi_3 = t^2$ on $[0, 1]$, nothing changes except how you compute inner products — for functions they are integrals:

$$\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt.$$

So the matrix entries are numbers like

$$\langle \varphi_1, \varphi_1 \rangle = \int_0^1 1 dt = 1, \quad \langle \varphi_2, \varphi_1 \rangle = \int_0^1 t dt = \frac{1}{2}, \quad \langle \varphi_3, \varphi_2 \rangle = \int_0^1 t^3 dt = \frac{1}{4},$$

and the right side is $\langle u, 1 \rangle$, $\langle u, t \rangle$, $\langle u, t^2 \rangle$ (integrals of u against each basis function). Solve the system, combine the φ 's with those coefficients, and that is the projection — no Gram-Schmidt needed.

3 Definitions and notation

Definition 3.1 (First definition). Write the definition and one sentence explaining why it matters.

4 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

5 Examples and calculations

Example 5.1 (Lecture example). Write the setup, the calculation, and the conclusion.

6 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: